Differentiation $f(x)$ is derivative x_0, if and only if

$$f'(x_0) = \lim_{\Delta x \to 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$$

E.g $f(x) = |x|$

① $f(x) = c$ $\quad f'(x) \equiv 0$

② $[c f(x)]' = c \cdot f'(x)$

③ $[f(x) + g(x)]' = f'(x) + g'(x)$

④ $[f(x) g(x)]' = f'(x) g(x) + f(x)' g(x)$

⑤ $\left(\frac{f(x)}{g(x)}\right)' = \frac{f'(x) \cdot g(x) - g(x)' \cdot f(x)}{[g(x)]^2}$

Chain rule

Lem (Alternative criterion for differentiability at a point):
A function $f: (a,b) \to \mathbb{R}$ is differentiable at $x_0 \in (a,b)$ if and only if there exists a function $F: (a,b) \to \mathbb{R}$

$$f(x) = f(x_0) + (x - x_0) \cdot F(x) \qquad ①$$

① $F(x)$ is continuous at x_0.

② If $F(x)$ is differentiable at x_0, then

$$F(x) = \begin{cases} \frac{f(x) - f(x_0)}{x - x_0} & \text{if } x \neq x_0 \\ \lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0} & \text{if } x = x_0 \end{cases}$$

Proof: if $x \in (a,b) \setminus \{x_0\}$ then since $F(x) = \frac{f(x) - f(x_0)}{x - x_0}$

$$f(x_0) + (x - x_0) \cdot F(x) = f(x_0) + \left(f(x) - f(x_0)\right) = f(x)$$

then ① holds

To verify $F(x)$ is continuous at x_0. We have $F(x) \to f'(x_0)$ as $x \to x_0$ and ① hold as $x \to x_0$.

Thm (Chain rule) Suppose $f:(a,b)\to(c,d)$ is differentiable at $x_0\in(a,b)$, and $g:(c,d)\to\mathbb{R}$ is differentiable at $f(x_0)$. Then

$$(g\circ f)'(x_0) = g'(f(x_0))\cdot f'(x_0)$$

$$\frac{dg}{dx} = \frac{dg}{df}\cdot\frac{df}{dx}$$

Proof: Since g is differentiable at x_0.

$$g(y) = g(f(x_0)) + (y - f(x_0))\,G(y)$$

then $G:(c,d)\to\mathbb{R}$ exists and $G(f(x_0)) = g'(f(x_0))$ $\quad y = f(x_0+\Delta x)$

$$g(f(x_0+\Delta x)) = g(f(x_0)) + (f(x_0+\Delta x) - f(x_0))\,G(f(x_0+\Delta x))$$

$$g(f(x_0+\Delta x)) - g(f(x_0)) = (f(x_0+\Delta x) - f(x_0))\,G(f(x_0+\Delta x))$$

$$\frac{g(f(x_0+\Delta x)) - g(f(x_0))}{\Delta x} = \frac{(f(x_0+\Delta x) - f(x_0))}{\Delta x}\,G(f(x_0+\Delta x))$$

Since $f(x)$ is differentiable at x_0. Let $\Delta x\to 0$

$$\lim_{\Delta x\to 0}\frac{g(f(x_0+\Delta x)) - g(f(x_0))}{\Delta x} = \lim_{\Delta x\to 0}\frac{(f(x_0+\Delta x) - f(x_0))}{\Delta x}\,G(f(x_0+\Delta x))$$

$$(g\circ f)'(x_0) = f'(x_0)\,G(f(x_0))$$

E.g $\quad [g(f(x_0))]' = \frac{dg}{dx} = \frac{df}{dx}\frac{dg}{df}$

$$[h(g(f(x_0)))]' = \frac{dh}{dx} = \frac{dh}{dg}\cdot\frac{dg}{df}\cdot\frac{df}{dx}$$

E.g $\quad f(x) = a^x \qquad a^x = \exp(\ln a^x) = \exp(x\cdot\ln a) \qquad y = x\cdot\ln a$

$$a^x = \exp(y)\cdot\ln a = \exp(x\cdot\ln a)\cdot\ln a = a^x\ln a$$

E.g. $\quad y = \sin(x^2) \qquad\qquad y = \frac{1}{\sqrt[3]{x^2+x+1}}$

$y = (\sin x)^2 \qquad\qquad y = (2x+1)^5(x^3-2x+3)^3$

$y = \sin(x^2)$ $\quad f(x) = x^2 \quad g(f) = \sin f$

$$\frac{dy}{dx} = \frac{dg}{df} \cdot \frac{df}{dx} = \cos f \cdot 2x = \cos(x^2) \cdot 2x$$

$y = \sin^2 x$ $\quad f(x) = \sin x \quad g(f) = f^2$

$$\frac{dy}{dx} = \frac{dg}{df} \cdot \frac{df}{dx} = 2f \cdot \cos x = 2\sin x \cdot \cos x$$

$y = \dfrac{1}{\sqrt[3]{x^2+x+1}}$ $\quad f(x) = x^2+x+1 \quad g(f) = (f)^{-\frac{1}{3}}$

$$\frac{dy}{dx} = (2x+1)\left(-\frac{1}{3}\right)\cdot(f)^{-\frac{4}{3}} = -\frac{1}{3}(2x+1)\left(x^2+x+1\right)$$

$y = (2x+1)^5(x^3-2x+3)^3$ $\quad g_1(x) = (2x+1)^5 \quad g_2(x) = (x^3-2x+3)^3$

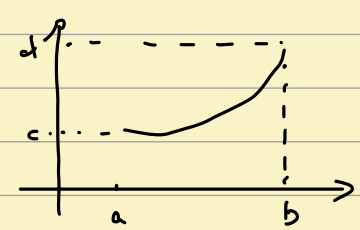
$f_1(x) = 2x+1 \quad g_1(f_1) = f_1^5 \quad g_1'(x) = 2\cdot 5(2x+1)^4$
$= 10(2x+1)^4$

$f_2(x) = x^3-2x+3 \quad g_2(f_2) = f_2^3 \quad g_2'(x) = (3x^2-2)\cdot 3(x^3-2x+3)^2$

$y = g_1(x)\cdot g_2(x) \quad y' = g_1'(x)\cdot g_2(x) + g_1(x)\cdot g_2'(x)$

$$= 10(2x+1)^4(x^3-2x+3)^3 + 3(2x+1)^5(3x^2-2)(x^3-2x+3)^2$$

Thm: Suppose that $f: [a,b] \to [c,d]$ is continuous and strictly increasing (decreasing), that $f(a) = c$ ($f(a) = d$), $f(b) = d$ ($f(b) = c$). Then the inverse function $f^{-1}: [c,d] \to [a,b]$ exists, is continuous, strictly increasing (decreasing) and surjective.



Thm: Suppose $f:[a,b]\to[c,d]$ is continuous and strictly increasing. and $x_0\in(a,b)$, $y_0=f(x_0)$. If f is differentiable at x_0, and $f'(x_0)\neq 0$. then f^{-1} is differentiable at y_0. and

$$(f^{-1})'(y_0)=\frac{1}{f'(x_0)} \qquad \frac{dx}{dy}=\frac{1}{\frac{dy}{dx}}$$

Proof: Take $x\in(x_0-\varepsilon, x_0+\varepsilon)$, $\varepsilon>0$. Write $y=f(x)$, $y_0=f(x_0)$

$\Rightarrow x=f^{-1}(y)$ and $x_0=f^{-1}(y_0)$

Since f is differentiable at x_0

$$\frac{f(x)-f(x_0)}{x-x_0}\to f'(x_0) \quad \text{as } x\to x_0$$

$$\Rightarrow \frac{y-y_0}{f^{-1}(y)-f^{-1}(y_0)}\to f'(x_0) \quad \text{as } x\to x_0$$

$$\Rightarrow \frac{f^{-1}(y)-f^{-1}(y_0)}{y-y_0}\to\frac{1}{f'(x_0)} \quad \text{as } x\to x_0$$

Since f is continuous at x_0, $y\to y_0$ as $x\to x_0$

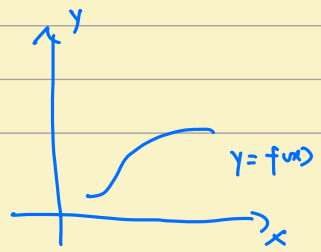
$$\Rightarrow \frac{f^{-1}(y)-f^{-1}(y_0)}{y-y_0}\to\frac{1}{f'(x_0)} \quad \text{as } y\to y_0$$

$$\Rightarrow [f^{-1}(y)]'=\frac{1}{f'(x_0)}$$

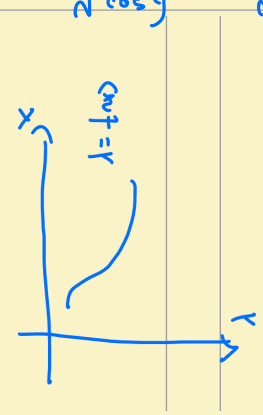
E.g. $y=\sin x \Rightarrow x=\arcsin y$

$y=f(x)=\arcsin x \qquad x=g(y)=\sin y$

$$f'(x)=\frac{1}{g'(y)}=\frac{1}{\cos y}=\frac{1}{\sqrt{\cos^2 y}}=\frac{1}{\sqrt{1-\sin^2 y}}=\frac{1}{\sqrt{1-x^2}}$$



$y=\arcsin x$



$y = f(x) = \arctan x \qquad x = g(y) = \tan y \qquad (\tan y)' = \sec^2 x$

$$= \frac{1}{\cos^2 x}$$

$$= \frac{\sin^2 x + \cos^2 x}{\cos^2 x}$$

$$= \tan^2 x + 1$$

$$f'(x) = \frac{1}{g'(y)} = \frac{1}{\sec^2 y} = \frac{1}{\tan^2 y + 1} = \frac{1}{x^2+1}$$

隐含函数

Implicit differentiation and logarithmic differentiation

$y = \sin^2 x \qquad y = \frac{1}{\sqrt[3]{x^3+1}}$

$x^2 + y^2 = 25 \Rightarrow y = \pm\sqrt{25 - x^2}$

$x^2 + 2xy + y^2 = 10$

E.g $\quad x^3 + y^3 = 6xy \Rightarrow x^3 + [f(x)]^3 = 6x \cdot [f(x)]$

$$g(x) = x^3 + [f(x)]^3 = 3x^2 + 3[f(x)]^2 \cdot f'(x) = 3x^2 + 3y^2 \cdot y'$$

$$6x[f(x)] = 6f(x) + 6xf'(x) = 6y + 6xy'$$

$$x^2 + y^2 \cdot y' = 2y + 2xy'$$

$$(y^2 - 2x)y' = -x^2 + 2y \Rightarrow y' = \frac{2y - x^2}{y^2 - 2x}$$

$x = 3 \Rightarrow x^3 + y^3 = 6xy \Rightarrow y = 3. \qquad y' = -1$

E.g. Find y' if $\sin(x+y) = y^2 \cos x$

$$[\sin(x+y)]' = (1+y')\cos(x+y) \qquad (f(x)^2)' = [f(x)]^2 f'(x)$$

$$[y^2 \cos x]' = 2yy'\cos x - y^2 \sin x \qquad = y^2 \cdot y'$$

$$(1+y')\cos(x+y) = 2yy'\cos x - y^2 \sin x$$

$$[-2y\cos x + \cos(x+y)]y' = -\cos(x+y) - y^2 \sin x$$

$$y' = \frac{y^2 \sin x + \cos(x+y)}{2y\cos x - \cos(x+y)}$$

Logarithmic differentiation

E.g. $y = \dfrac{x^{\frac{3}{4}}\sqrt{x^2+1}}{(3x+2)^5}$

$$\ln y = \frac{3}{4}\ln x + \frac{1}{2}\ln(x^2+1) - 5\ln(3x+2)$$

$$(\ln y)' = \frac{1}{y}\cdot y' = \frac{3}{4x} + \frac{1}{2}\cdot\frac{2x}{x^2+1} - 5\cdot\frac{3}{3x+2}$$

$$y' = \frac{x^{\frac{3}{4}}\sqrt{x^2+1}}{(3x+2)^5}\left(\frac{3}{4x} + \frac{1}{2}\cdot\frac{2x}{x^2+1} - 5\cdot\frac{3}{3x+2}\right)$$

E.g $y = x^{\sqrt{x}}$

$$\ln y = \sqrt{x}\cdot\ln x \qquad \frac{1}{y}y' = \frac{1}{\sqrt{x}} + \ln x\cdot\frac{1}{2\sqrt{x}}$$

$$y' = x^{\sqrt{x}}\left(\frac{2+\ln x}{2\sqrt{x}}\right)$$

Rolle's theorem

Thm (Rolle's theorem): Suppose f is a function satisfy the following:

① f is continuous on the closed interval $[a,b]$,

② f is differentiable on the open interval (a,b).

③ $f(a) = f(b)$

Then, there is a number $c \in (a,b)$, such that $f'(c) = 0$.

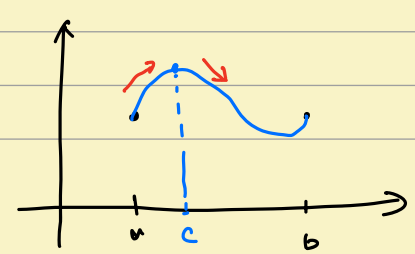
Proof: By the minimum-maximum theorem, f is continuous on $[a,b]$, then

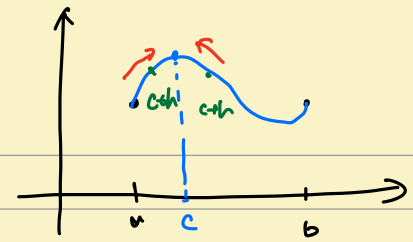
$M = \max f(x) \qquad m = \min f(x)$

Case ① $f(a) = f(b) < M$, or

Case ② $m < f(a) = f(b)$,

Case ③ $m = f(a) = f(b) = M$





when $h<0$ $h\to 0$ $f(c+h)-f(c)<0$

when $h>0$ $h\to 0$ $f(c+h)-f(c)>0$

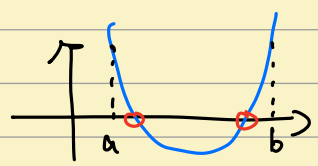
$$\lim_{h\to 0^-}\frac{f(c+h)-f(c)}{h}\le 0 \qquad \lim_{h\to 0^+}\frac{f(c+h)-f(c)}{h}\ge 0$$

since f is differentiable $c\in(a,b)$

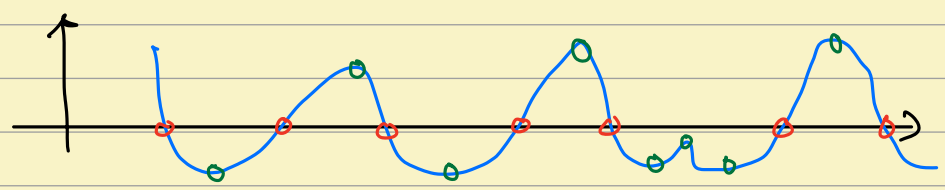
$$f'(c)=\lim_{h\to 0}\frac{f(c+h)-f(c)}{h}=\lim_{h\to 0^-}\frac{f(c+h)-f(c)}{h}$$

$$=\lim_{h\to 0^+}\frac{f(c+h)-f(c)}{h}=0$$

Corollary: If $f:(a,b)\to\mathbb{R}$ is differentiable, then between any two different zeros of, there is at least a zero of f'.



Corollary: $f:(a,b)\to\mathbb{R}$ is differentiable, and has n distinct zeros, then f' has at least $n-1$ differentiable.

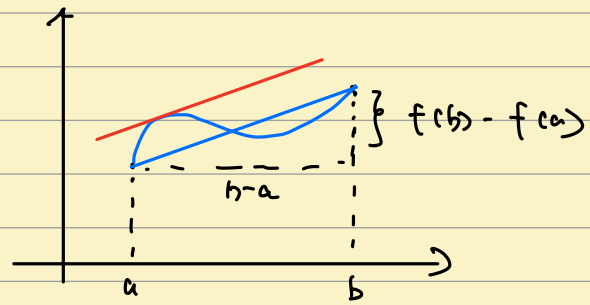


Equivalently, if f' has m distinct zeros, the f has at most $m+1$ distinct zeros.

Thm (Mean Value Theorem). Suppose f is a function that satisfies

① f is continuous at the closed interval $[a, b]$,

② f is differentiable at the open interval (a, b)

Then, there exists a number $c \in (a, b)$, such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$